PROJECT PROPOSAL: DSA 210 Introduction to Data Science

Project Title: The Sound of Happiness: A Cross-National Analysis of Musical Genre Preferences and Subjective Well-being

1. Motivation and Problem Definition The relationship between cultural consumption and psychological well-being is a key area of study in behavioral science. As an Industrial Engineering student, I aim to investigate whether the dominant musical genres in a country can serve as a predictor for its national happiness levels. This project will explore if "happier" nations consume different types of music (e.g., high-energy vs. acoustic) compared to nations with lower happiness scores, moving beyond simple demographics to cultural indicators and whether economic factors of a country like inflation and GDP have an effect on music taste of that country.

2. Data Source and Collection Method

- **Happiness Data: The "World Happiness Report" dataset (annual happiness scores, GDP, and social support by country) will be retrieved from Kaggle.**
- **Music Data: The "Spotify Global Top Charts" or "Music Genre by Country" dataset, which includes popular genres and audio features (Valence, Energy, Danceability) per region, will be sourced from Kaggle.**
- **Economic Data: The inflation rate data will be retrieved from Kaggle.**
- **Collection Method: Both datasets will be downloaded in CSV format and merged using Python's pandas library based on the "Country" and "Year" columns.**

3. Characteristics of the Data

- **Time Period: 2020 to 2026.**
- **Sample Size: Approximately 150 countries with multi-year observations.**
- **Features: Happiness Score, GDP per capita, Primary Genre, and Audio Features (Valence, Energy, Tempo).**